

Eco-friendly strategic decision-making towards grid modernization with fully renewable water and power community hubs

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ABSTRACT

By growing the energy generation tendency towards 100% renewable systems in modern energy grids, optimal decision-making has become a fundamental challenge due to the uncontrollable volatilities of renewable energy sources, making their optimal integration hardly feasible. Due to the undeniable dependencies between the water and power sectors, this paper proposes an innovative adaptive robust reinforcement learning decision-making model for the operation of community energy hubs with fully renewable energy sources in the water and power integrated network. The proposed eco-friendly model benefits the reinforcement learning technique for detecting an optimal decision-making strategy for energy hubs as well as empowers both robust and stochastic programming methods for purposeful modeling uncertainties in the system. Given the equipping of energy hubs with 100% renewables and the importance of flexibility in their interactions, a novel twofold ancillary service framework is developed to allow energy hubs to simultaneously benefit transactive energy management and demand response schemes. After conducting a simulation on the modified IEEE 33-bus test system, the results denoted a 17.03% reduction in the total cost when all hubs are operated under the proposed model than the traditional one. Moreover, the results assessment discovered that as uncertainty quantification gives a realistic overview to the decision-maker for making purposeful decisions, it imposes 8.96% more costs in comparison with the basic state of the system without uncertainty modeling. In addition to economic advantages, satisfying technical and environmental goals proved the effectiveness of the suggested model in optimally integrating renewables fully into the interconnected energy structure.

Nomenclature

Indexes	
i, j	Index for buses in the power grid (PG)
t	Index for daily hours
h	Index for energy hubs
Parameters	
ρ_t^{WS}, ρ_t^{EL}	Water and power selling prices for consumers, ((\$/m ³) and (\$/kW))
ϕ_h^{RSU}	Degradation cost of the battery storage unit (BSU), (\$)
$\eta_h^{Le,B}, \eta_h^{CD,B}$	Leakage and charging/discharging loss factors
$\rho_t^{W,Ex}, \rho_t^{E,Ex}$	Water and electricity exchange prices, ((\$/m ³) and (\$/kW))

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$W_{h,t}^{De}, P_{h,t}^{De}$	Water and power demands, ((m ³) and (kW))
$\eta_h^{FC}, \eta_h^{ELC}$	Fuel cell (FC) and electrolyzer (EL) efficiencies
σ^{Hy}	Lower heating coefficient of the hydrogen
$\lambda^{H2}, \epsilon^{Gas}$	Gas volume (m ³) and constant for the hydrogen storage unit (HSU)
$P_h^{WS,Ra}, v_h^{Ra}$	Rated wind power and speed for energy hub h , ((kW) and (m/s))
v_t, ω_t^{SS}	Wind velocity and solar radiation at time t , ((m/s) and (kW/m ²))
$v_h^{C,O}, v_h^{C,I}$	Cut-out and cut-in wind speeds, (m/s)
$\Omega_a^{WS}, \Omega_b^{WS}, \Omega_c^{WS}$	Coefficients for computing wind power production
N_h^{SS}, η^{SS}	Number of solar cells and their efficiency
δ^{SS}, δ^{SS}	Coefficients for computing solar power production

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τ^M, τ^A	Mean and ambient temperatures (°K)
$N_{h,BSU,LC}$	Life cycle number of the BSU
$C_{h,BUS,IN}, g_{h,BSU,Ra}$	Investment cost of the BSU and its rated energy (\$)
$Y_{i,j}$	Admittance of the line i - j in the PG
$\gamma_{h,A,WS}, \eta_{h,WP}$	The altitude of the water storage unit (WSU) location (m) and pump efficiency
ψ^{Wa}, γ^{Wa}	Gravity (kg/m ³) and density of the water (m ³)
$\eta_{h,WD}, \gamma_{h,WW}$	The efficiency of the water desalination (WD) and water level of the water well (WW)
ϕ^{Wa}, ϕ^{EL}	Coefficients for the water and electricity sharing
$\theta_{h,ILP1}, \theta_{h,ILP2}$	Coefficients for modeling the curtailed load cost
Variables	
$P_{h,t}^{WUs}$	Total consumed power (kW) by the water network (WN)
$P_{h,t}^{BSU}, g_{h,t}^{BSU}$	Power (kW) and energy (kWh) changes for the BSU
$C_{h,t}^{SHP}, C_{h,t}^{ILP}$	Cost of the shifting and interrupted load programs (\$)
$W_{h,t}^{EX}, P_{h,t}^{EX}$	Water (m ³) and power exchanging (kW) with the main grid
$P_{h,t}^{ELC}, P_{h,t}^{FC}$	Consumed/produced electricity by the EL/FC (kW)
$P_{h,t}^{HM}, P_{h,t}^{MHI}$	The delivered power from the energy hub to the local power market (LPM) and vice versa (kW)
$P_{h,t}^{WS}, P_{h,t}^{SS}$	Outputs of wind and solar systems (WS and SS) (kW)
$\phi_{h,t}^{H2,FC}, \phi_{h,t}^{H2,ELC}$	Consumed/produced hydrogen by the FC and EL (mol)
$H_{h,t}^{HSU}$	The amount of hydrogen stored in the HSU (mol)
$o_{h,t}^{BSU}$	The state of charge of the BSU
$V_{i,t}, \theta_{i,t}$	The PG's bus voltage (V) and phase angle (rad)
$P_{i,t}^{Gen}, P_{i,t}^{Con}$	Active power generation and demand (kW)
$Q_{i,t}^{Gen}, Q_{i,t}^{Con}$	Reactive power generation and demand (kVAR)
$W_{h,t}^{WSCh}, W_{h,t}^{WSDis}$	Charging and discharging of the water in the WSU (m ³)
$W_{h,t}^{HM}, W_{h,t}^{MHI}$	The delivered water from the energy hub to the local water market (LWM) and vice versa (m ³)
$W_{h,t}^{ELC}$	The amount of consumed water by the EL (m ³)
$W_{h,t}^{WWDis}, W_{h,t}^{WD}$	Produced water by the WW and WD (m ³)
$P_{h,t}^{WWDis}, P_{h,t}^{PWD}$	Consumed power by the WW and WD (kW)
$P_{h,t}^{WSU,Ch}, \gamma_{h,t}^{WS}$	Consumed power by the WSU (kW) and its water level (m ³)
$\chi_{h,t}^{WSCh}, \chi_{h,t}^{WSDis}$	Binary variables for charging and discharging of the WSU
$\chi_{h,t}^{WM}, \chi_{h,t}^{WMHI}$	Binary variable for the water sharing in the LWM
$\chi_{h,t}^{LPM}, \chi_{h,t}^{LPMHI}$	Binary variable for the power-sharing in the LPM
$\theta_{h,t}^{ILP}, \theta_{h,t}^{SHP}$	The amount of curtailed and shifted loads (kW)

1. Introduction

1.1. Motivation

Recently, the rapid growth in energy demand under the significant increase in global population has been accompanied by the escalating accumulation of greenhouse gas pollution leading to becoming the energy transition an inevitable trend of grid expansion [1]. This transformation in the energy landscape is intensified, particularly by recent progress in digitalization, decentralization, decarbonization, democratization, and urbanization [2]. Such developments have also highlighted undeniable dependencies among energy networks and necessitate the need for interdependently optimizing energy networks instead of separately modeling them. In this context, recent advancements in energy conversion and storage technologies, as well as water-energy nexus models, reveal irrefutable reliances between water and power grids [3]. To reach the decarbonization goals of modern energy networks, the integrated exploitation of water and power grids with 100% renewable energy sources (RESs) is intended as one of the overarching goals in line with zero-carbon emissions plans. However, such a structure imposes huge uncertainties making optimal decision-making a great challenge in the stochastic environment. RESs offer different technologies for harvesting clean energy in an environmentally friendly manner. In this work, the coupled power and water structure benefits solar panels and wind turbines as two widely used types of RESs in generating eco-friendly power and their output's dependency on climate variations has led to imposing uncertainties to the power production sector. On the

other hand, energy price also varies at different hours of the day due to diverse patterns of power consumption in various periods of the day. Hence, effectively modeling such uncertainties can assist the system in gaining the most confident results in the deregulated environment. Due to this, wind speed, solar irradiation, and power prices are accommodated as uncertain sources for this research that are taken into account in the uncertainty quantification process. For such an energy structure under the high penetration of different uncertainties, the current research work proposes an innovative adaptive robust reinforcement learning (ARRL) decision-making model to detect an optimal decision-making strategy for operating community energy hubs with 100% RESs in the interconnected water and power grid (IWPG).

1.2. Literature survey

In recent studies, different endeavors have been carried out to disclose the key linkages between the water and power sectors by developing hybrid frameworks under diverse optimization models. Such power and water interactions are modeled in Ref. [4] by proposing a two-stage operation model that benefits the distributionally robust optimization for capturing the wind uncertainty in the co-optimization of interconnected water and power systems. A cyber-secured operation of water and power systems is analyzed in Ref. [5] by offering a two-stage risk-averse strategy with the aim of obtaining economic objectives. From a multi-energy perspective, a robust operation of water-energy nexus systems is taken into account in Ref. [6] to explore interdependences among multi-carrier energy devices in the integrated energy system. The uncertainty-informed exploitation of water-energy nexus systems is expanded in Ref. [7] by adopting a joint probabilistic constraint programming method enabling the probabilistic modeling of IWPGs. This is while jointly managing energy and water systems is addressed in Ref. [8] by suggesting a simulating methodology for the integrated water and energy system aiming to quantify multifarious modeling assumptions. On the other hand, the power and water interaction is modeled in Ref. [9] by proposing a short-term wind-hydrothermal exploitation framework that uses pumped-hydro storage to support the energy supply of the renewable-based system aiming to minimize energy costs.

Despite modern energy networks are planned for fully utilizing RESs in the grid modernization process, developing thorough operation models is overlooked for IWPGs in line with zero-carbon emissions programs. Herein, as substantial developments in machine learning techniques can unlock the challenge of detecting the optimal decision-making strategy for renewable-based IWPGs, their effective applications are not intended in the coupled water and power domain. This is while recent literature clarifies the widespread application of reinforcement learning (RL) methods in different energy problems. For example, a multi-agent RL method is proposed in Ref. [10] to allow microgrids to learn the optimal policies in the scheduling of the renewable-based system. In order to control voltage volatilities and coordinate active and reactive power in the presence of distributed RESs, the authors proposed a multi-agent RL methodology in Ref. [11] to alleviate voltage fluctuations and decrease power loss. This is while the adaptive voltage regulation is carried out in Ref. [12] by suggesting a safe deep RL to achieve minimum energy costs considering the intermitteances of the energy price and load as well as RESs. To decline the negative impact of stochastic fluctuations associated with the mentioned uncertain parameters in the optimal scheduling of isolated microgrids, an automated RL-based framework is proposed in Ref. [13] considering economic objectives. In Ref. [14], the authors developed an optimization embedded RL to make the achievement of adaptive decision-making possible for automated vehicles. A multi-agent framework is presented in Ref. [15] based on the deep RL to the economic and decentralized frequency control of networked-microgrid systems. The application of RL is expanded in Ref. [16] for developing the demand response algorithm to manage energy with the aim of retaining a satisfaction index as

well as minimizing electricity costs. A deep RL methodology is applied in Ref. [17] to develop a three-stage optimization strategy for the optimal operation of microgrids by focusing on information privacy preserving of them. As the mentioned studies give the applicable overview of RL-based paradigms for improving various energy processes, all of them suffer from being adequately robust against uncontrollable variations of RESs highlighting deficiencies of RL in making energy structures with a high contribution of renewables enough robust. For instance, although a multi-agent broad RL algorithm is proposed in Ref. [18] to adopt the broad network to intelligently control traffic light, its framework does not support the system's robustness against multifarious uncertainties. The authors in Ref. [19] implemented deep RL algorithms for the optimal economic dispatch of gas turbines to benefit the incorporation of the environment model and thermodynamic software in meeting the system's goals. In another work [20], the authors proposed a fuzzy multi-objective RL to present an effective solving methodology for a bi-objective reach-avoid game based on the fuzzy Q-learning algorithm. Moreover, a policy gradient RL method is presented in Ref. [21] to use fuzzy information for dynamically controlling of biped walking of humanoid robots. However, as the aforementioned works have focused on developing applicable models of RL for addressing various types of challenges, all of them also highlight the deficiencies of RL-based approaches in rendering robust conditions for the secure operation of the system in the presence of numerous renewables. Due to this, the current article develops an ARRL model that benefits the advantages of both RL and hybrid robust stochastic (HRS) technique to not only effectively gain optimal set points for energy units but also provide sufficient robustness for the renewable-dominant community energy hubs.

Albeit the aforementioned studies render effective machine learning-based solutions for managing and controlling energy in the energy structure with different challenges, they have not addressed the energy management of the system with 100% RESs in the IWPG structure. This is while increasing the share of RESs towards a 100% level in the power production sector of future modern energy grids has highlighted the necessity of developing the most holistic models for facilitating the adoption of 100% RESs [22]. Given the RESs penetration is going to be 100%, a two-stage operation strategy is proposed in Ref. [23] along with developing a new market framework for 100% RESs. A novel modeling framework for the power system with 100% renewables is also proposed in Ref. [24] by using a decomposition-based solution strategy for generation and transmission expansion planning. Moreover, a multiyear expansion-planning strategy is suggested in Ref. [25] to scrutinize the effectiveness of interconnected islands in promoting 100% RESs as well as minimizing the total cost of the system. The planning of energy systems is also taken into account in Ref. [26] by proposing a multi-objective optimal planning strategy for renewable-penetrated energy hubs to simultaneously reduce environmental pollution and energy costs. In another work [27], a multi-objective strategy is designed for the energy management of a virtual power plant with different renewables for pursuing various objectives including voltage security, operational, and economic in the distribution system. Another new energy management strategy is manipulated in Ref. [28] for the virtual power plant in which the network-constrained unit commitment has been done based on the hybrid flexible-securable exploitation model to improve economic achievements. In Ref. [29], a comprehensive energy management strategy is proposed for the smart distribution grid to evaluate some critical goals such as reliability, flexibility, operation, and economics by considering different uncertainties based on the scenario-dependent method. Moreover, an optimal planning strategy is used in Ref. [30] for the energy management of the hybrid energy system to supply both electrical and thermal energy by prioritizing the utilization of renewables to realize cost minimization and energy efficiency improvement. This is while developing a new operation strategy is accommodated as another prominent step for the optimal energy management of energy hubs that has attracted more attention in recent literature. For instance, the authors in Ref. [31] analyzed the optimal

decision-making process for energy hubs by designing a new operation strategy under the scenario-based modeling of various uncertainties. Additionally, a robust chance-constrained methodology is taken into account in Ref. [32] to effectively model intermittences of uncertain sources with the aim of maximizing loadability index while properly addressing the risk of decision-making in the presence of renewables. Table 1 clearly presents the innovative features of this work in comparison with recent literature.

1.3. Research gaps

Albeit the recent literature contains different water-energy nexus paradigms to model the variety of interactions between water and multi-carrier energies, several basic shortcomings have led to remaining critical challenges in modernizing IWPGs.

- Although future modern energy grids aim to utilize 100% RESs for clean energy generation in their energy production sector, developing a viable model is not intended for IWPGs yet in accordance with zero-carbon emission plans. This is while effective interactions between water and power systems indicate the promising possibility of integrating 100% RESs in IWPGs.
- The optimal operation of renewable-dominant energy hubs requires a holistic decision-making strategy to enable decision-makers for optimal actions in the presence of 100% RESs. However, none of the recent studies have paid attention to developing an innovative solution for finding the optimal decision-making strategy for community energy hubs to enable them for reliable exploitation in the structure with 100% RESs.
- Energy structures with 100% RESs need capable ancillary services to reliably benefit fully carbon-free produced energy, flexibly act in energy interactions, and continuously keep the sustainability of the grid. However, developing such ancillary services has been ignored in recent works leading to remaining the challenge of achieving the mentioned advantages for the IWPG.

1.4. Contributions

Given the expressed research gaps in the previous subsection, this work proposes an innovative ARRL decision-making strategy to provide realistic conditions along with confident results for the decision-maker to make purposeful decisions in the deregulated environment. The proposed model is model-free that provides appropriate conditions for exploring the system with the need for more sophisticated information. It also carries an off-policy feature that enables the framework to detect the optimal policy by intending all limitations of hybrid units and grids in line with realizing the defined objectives. It pursues the policy detection process for the optimal operation of hubs by learning the most effective steps in improving the objective function considering the worst states of uncertainties in the studied environment. In addition to discovering the optimal policy, it procures a sufficient degree of robustness and realizes techno-economic-environmental objectives for community energy hubs in the system with 100% RESs. The research gaps mentioned in the previous subsection have motivated us to propose practical solutions for filling them by offering the following contributions.

- In response to the lack of a holistic model for optimal integration of 100% RESs in IWPGs structure, this article develops a capable model of operation for community energy hubs allowing them to produce fully carbon-free power in the renewable-dominant structure of IWPGs. The proposed model is empowered by energy conversion and storage facilities along with energy trading and management schemes for the secure water and power supply.
- Given the challenge of applying a perfect decision-making strategy for IWPGs with 100% RESs, this work develops an innovative model-

Table 1

Innovative features of this research than recent literature.

Ref.	Ancillary service framework	Decision-making strategy	100% RESs	Uncertainty modeling approach	IWPG structure	
					Power	Water
[4]	×	Two-stage scheduling strategy	×	Distributionally robust optimization	✓	✓
[10]	Bidding-based incentive mechanism	Multi-agent reinforcement learning	×	×	✓	×
[14]	×	Embedded reinforcement learning	×	×	✓	×
[31]	×	Optimal operation strategy	×	Scenario-based method	✓	×
[5]	×	Two-stage risk-averse strategy	×	Conditional value-at-risk (CVaR)	✓	✓
[32]	×	Optimal operation strategy	×	Robust chance-constrained	✓	×
[25]	×	Multiyear expansion-planning	✓	×	✓	×
[26]	Demand response programs	Multi-objective optimal planning strategy	×	Return on assets methodology	✓	×
[15]	×	Multi-agent reinforcement learning	×	×	✓	×
[27]	Price-based demand response	Fuzzy decision-making	×	Unscented transformation method	✓	×
[6]	×	Two-stage operation strategy	×	Robust optimization (RO)	✓	✓
[13]	Demand response program	Automated reinforcement learning	×	Chance constraint programming	✓	×
[7]	×	Day-ahead operation strategy	×	Joint probabilistic constraint programming	✓	✓
[23]	×	Two-stage operation strategy	✓	×	✓	×
[28]	Demand response	Two-layer coordinated strategy	×	Stochastic programming	✓	×
[8]	×	Customisable water-energy simulation strategy	×	Scenario-based method	✓	✓
[24]	×	Decomposition-based solution	✓	Stochastic programming and RO	✓	×
[29]	×	Energy management strategy	×	Scenario-based stochastic optimization	✓	×
[30]	×	Optimal planning strategy	×	Point estimate method	✓	×
[33]	Demand response program	Deep reinforcement learning	×	×	✓	×
This paper	Transactive energy-based TAS framework	Adaptive robust reinforcement learning strategy	✓	HRS technique	✓	✓

free and off-policy ARRL decision-making strategy that enables the decision-maker to adopt purposeful decisions against the worst stochastic violations of uncertain parameters. The proposed paradigm is novel for its capability in learning the optimal decision-making strategy considering the worst states of uncertainties to procure a sufficient degree of robustness as well as achieve techno-economic-environmental objectives for community energy hubs in the system with 100% RESs.

- To address the challenge of reliable water and power supply in the presence of 100% RESs, this article proposes a twofold ancillary service (TAS) framework that simultaneously benefits transactive energy technology and demand response programs to provide confident ancillary services. The proposed framework enables community energy hubs to reliably serve water and power to consumers, flexibly act in multi-energy interactions, and effectively maintain the sustainability of the system.
- Optimally operating energy hubs is a critical challenge when the system faces a high level of stochastic variations coming from different uncertain sources. To effectively model uncontrollable features of various uncertain parameters with diverse stochastic behaviors, this work adopts a HRS technique to tackle intermittences of RESs and energy prices aiming to simultaneously bring satisfactory robustness and techno-environmental-economic advantages.

1.5. Paper organization

The rest of this article has the following organization. Section 2 elaborates on the eco-friendly structure of community energy hubs. Section 3 describes the problem formulation containing the uncertainty modeling process, operational constraints of water and power networks, the TAS framework, and the ARRL model. Section 4 is for simulation results by presenting case studies and discussing the results. Ultimately, Section 5 concludes this paper.

2. Eco-friendly structure for community energy hubs

One of the overarching aims of this work is to fulfill the critical gap regarding the existence of an eco-friendly structure for optimally integrating 100% RESs in IWPGs. This paper develops a water-energy nexus

structure for community energy hubs that welcome different types of RESs, energy storage and conversion facilities, and water and power managing programs to realize fully clean energy generation in accordance with the decarbonization plans of the grid modernization. Fig. 1 portrays the schematic of this structure in the IWPG.

As Fig. 1 shows, wind and solar systems (WSs and SSs) are RESs used for pollutant-free power production in energy hubs. Given the

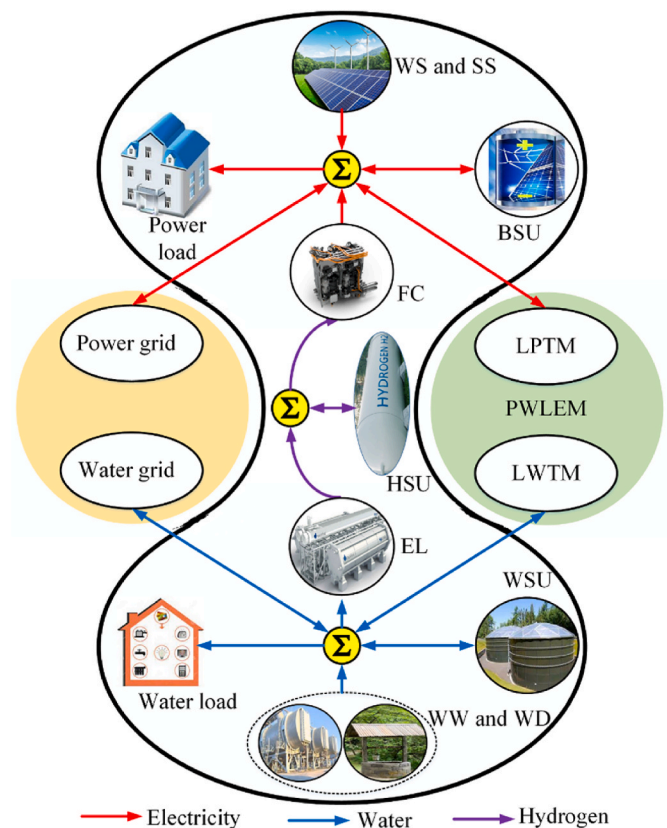


Fig. 1. Community energy hubs in the integrated power and water grid.

renewable-dominant structure requires supportive devices to keep energy balance, the battery storage unit (BSU) is operated to improve the power supply process by performing charge or discharge in different time periods. To increase the assurance of energy hubs' flexibility, the energy conversion is established by deploying the electrolyzer (EL) for producing the hydrogen molar, the hydrogen storage unit (HSU) for storing the hydrogen, and the fuel cell (FC) for the reproduction of power in the required time periods. Indeed, the excess of generated clean power from RESs in the power sector is used by the EL to produce hydrogen molar. Then, the produced hydrogen molar is stored in the HSU to provide the opportunity of re-generating electricity by the FC when the power grid faces a drop in power production by RESs. Indeed, when the power balance is going to be violated due to the uncertain decline in RESs' output, the stored hydrogen in the HSU can be used by the FC to generate more electricity to supply to the system to keep the power balance and assist in the sustainability maintenance of the renewable-based structure. Therefore, the hydrogen-based units are deployed aiming to create the energy conversion possibility to enhance the flexibility of the power grid, and therefore, the hydrogen network with a separate load and distribution system is not accommodated in this work.

In energy hubs, the EL's need for water to produce the hydrogen molar along with the water systems' need for power to be exploited in the IWPG is clear evidence of the dependencies between water and power sectors. The water sector benefits from water desalination (WD) and water well (WW) as the main sources of water generation. Similar to the power grid (PG), the water network (WN) exploits the water storage unit (WSU) to reliably meet the water load in the deregulated environment. The flexibility of energy hubs is further improved by employing transactive energy technology for constructing the power and water local energy market (PWLEM), in which each energy hub can freely share water and power with other hubs to achieve its various goals in the local water and power trading markets (LWTM and LPTM). Moreover, energy hubs are allowed to have limited water and power exchange with the WN and the PG due to the technical and economic objectives.

3. Materials and methods

3.1. Adaptive robust reinforcement learning (ARRL) model

In this study, the objective function is structured based on the minimization of the total cost for community energy hubs according to the following formula.

In (1), Θ_h states the objective function. The first two terms are for modeling the cost of power consumption in the WN and the operation cost for the BSU. The third term states the cost of implementing the TAS framework. The fourth and fifth terms are the revenue/cost of water and power exchange with the upstream network. The sixth and seventh terms model the revenue of the sold water and power to consumers. All required parameters and variables are provided in the nomenclature table. Given the basic version of the objective function without uncertainty modeling in (1), the deterministic model of the system can be defined as follows.

$$\begin{aligned} \text{Min } \Theta_h \\ \text{s.t.} \end{aligned}$$

All constraints (11) to (49)

As mentioned, the optimal operation of energy hubs with 100% RESs is engaged with diverse types of uncertainties that affect the realistic modeling of energy systems. In such a condition, developing a hybrid model is crucial in uncertainty modeling not only for procuring adequate robustness but also for resolving the high stochastic volatilities of RESs [34]. Indeed, in this work, the full potential of RESs is used to produce 100% clean power in community energy hubs. As the generation sector is fully equipped to RESs as stochastic producers, power production faces a high level of uncertainties making the optimal scheduling of the system and reliable energy supply a great challenge. In such a deregulated environment, making purposeful decisions is another key challenge for the decision-maker. Herein, robust optimization can effectively consider the worst state of uncertain changes to provide a sufficient degree of robustness for the renewable-based system enabling a decision-maker to make confident decisions. However, as robust optimization can procure a sufficient degree of robustness for community hubs by considering the worst state of uncertainties' fluctuations, finding optimal set points for different energy systems is still an important challenge for the optimal operation of energy structure. To simultaneously utilize the advantages of both robust optimization and RL, an ARRL decision-making model is developed to detect an optimal decision-making strategy for operating community energy hubs with 100% RESs in the IWPG. In the first step of the proposed ARRL model, robust optimization is exerted to give a robust framework by modeling the intermittences of the energy price. Applying the robust framework converts the problem formulation as follows.

$$\begin{aligned} \min \Theta_h = \mathbb{R}_h^{\Theta_1} - \mathbb{N}_h^{\Theta_2} \quad \forall h \\ \text{s.t.} \end{aligned}$$

$$\begin{aligned} \Theta_h = \sum_{t=1}^T \rho_t^{EL,S} P_{h,t}^{WUs} + \sum_{t=1}^T \left[\left(\phi_h^{BSU} \eta_h^{Le,B} g_{h,t}^{BSU} \right) + \left(\phi_h^{BSU} P_{h,t}^{BSU} \right) \right] + \sum_{t=1}^T (C_t^{SHP} + C_t^{ILP}) - \sum_{t=1}^T W_{h,t}^{Ex} \rho_t^{WEx} \\ - \sum_{t=1}^T P_{h,t}^{Ex} \rho_t^{EEx} - \sum_{t=1}^{t_f} \rho_t^{W,S} W_{h,t}^{De} - \sum_{t=1}^T \sum_{i=1}^B \rho_t^{EL,S} P_{h,t}^{De} \end{aligned} \quad (1)$$

$$\begin{aligned} \mathbb{R}_h^{\Theta_1} = \sum_{t=1}^T \rho_t^{EL,S} P_{h,t}^{WUs} + \sum_{t=1}^T \left[\left(\phi_h^{BSU} \eta_h^{Le,B} g_{h,t}^{BSU} \right) + \left(\phi_h^{BSU} P_{h,t}^{BSU} \right) \right] + \sum_{t=1}^T (C_t^{SHP} + C_t^{ILP}) - \\ \sum_{t=1}^T W_{h,t}^{Ex} \rho_t^{WEx} - \sum_{t=1}^{t_f} \rho_t^{W,S} W_{h,t}^{De} - \sum_{t=1}^T \sum_{i=1}^B \rho_t^{EL,S} P_{h,t}^{De} \end{aligned} \quad (2)$$

$$\begin{cases} \mathbb{N}_h^{\Theta 2} \leq \left[-\sum_{t=1}^T P_{h,t}^{Ex} \cdot \rho_t^{E,Ex^*} - \min_{Y, \psi_t} \left(r\Xi + \sum_{t=1}^T \psi_t \right) \right], P_{h,t}^{Ex} \leq 0 \\ \mathbb{N}_h^{\Theta 2} \leq \left[\sum_{t=1}^T P_{h,t}^{Ex} \cdot \rho_t^{E,Ex^*} - \min_{Y, \psi_t} \left(r\Xi + \sum_{t=1}^T \psi_t \right) \right], P_{h,t}^{Ex} \geq 0 \end{cases} \quad (3)$$

$$\begin{cases} r + \psi_t \geq P_{h,t}^{Ex} \cdot \rho_t^{E,Ex^+}, P_{h,t}^{Ex} \leq 0 \\ r + \psi_t \geq P_{h,t}^{Ex} \cdot \rho_t^{E,Ex^-}, P_{h,t}^{Ex} \geq 0 \end{cases} \quad (4)$$

+ All constraints (10) to (49)

where, $\mathbb{R}_h^{\Theta 1}$, $\mathbb{N}_h^{\Theta 2}$ are auxiliary variables, Ξ is the budget of uncertainty, ξ_t is an auxiliary variable used for controlling the degree of robustness, ψ_t and r are dual variables used to formulate the dual problem for replacing it with a primal one as follows. This research develops an HRS method to deal with stochastic fluctuations with different behaviors. In this respect, the outputs of RESs are modeled using stochastic programming, in which the Latin hyperbolic sampling (LHS) method [35,36] is availed for generating scenarios whereas the fast forward selection (FFS) method [37,38] is considered for reducing the number of scenarios to give the advantages in terms of complexity, computational burden, and running time. Under the applied scenarios, the expected value of the objective function is given as:

$$\min \Theta_h = \sum_{s=1}^S \zeta_s \cdot \Theta_{h,s} \quad (5)$$

where, the probability of scenarios is stated by ζ_s .

In the second step of the proposed ARRL model, the RL is developed to learn the optimal decision strategy for the operation of community energy hubs. The detection of the optimal decision strategy in the RL-based approach is done based on interactions between the agent and environment, in which the Markov Decision Process (MDP) is used for presenting the environment [39,40]. During the process, the exploration and exploitation abilities allow the agent to learn the optimal strategy by interacting in different episodes [39]. In this respect, an agent receives a specific reward from the environment in response to its conducted action and the main objective is to find the optimal policy that gives the maximum cumulative reward as follows [41].

$$\pi^* = \arg \max_{\pi: \mathfrak{S} \rightarrow \Psi} E_{\pi} \left[\sum_{t=1}^{\infty} \gamma^{t-1} \cdot \mathfrak{R}_t \right] \quad (6)$$

where, π is the policy and π^* is its optimal amount. $\mathfrak{S}$ and Ψ are states and actions in RL. $\mathfrak{R}_t$ is the reward and γ states the discount factor applied for reducing the effect of future rewards. In RL, the state-value functions are typically availed for constructing the optimal policy as the estimations of the expected return in a specific time period. However, applying them to directly obtain the optimal policy is not easy due to their dependencies on the model information such as reward distributions and transition probabilities that are usually unavailable [41]. Thus, the Q-function or action-value function is intended as follows.

$$Q^{\pi}(s, a) := E_{\pi} \left[\sum_{t=1}^{\infty} \gamma^{t-1} \cdot \mathfrak{R}_t | S_0 = s, A_0 = a \right], S \in \mathfrak{S}, A \in \Psi \quad (7)$$

$$Q^*(s, a) := \max_{\pi: \mathfrak{S} \rightarrow \Psi} Q^{\pi}(s, a) \quad (8)$$

$$\pi^*(s) = \arg \max_{a \in \Psi} Q^*(s, a) \quad (9)$$

where, $Q^*(s, a)$ and $\pi^*(s)$ are the optimal amounts of the Q-function and policy. After computing $Q^*(s, a)$ in (8), the optimal policy can be obtained by (9), which is model-free without depending on the model

information. Based on the type of the model, policy, and value functions, RL methodologies are broadly categorized into several classes that can be fully accessed in Ref. [41]. Among multifarious methods proposed for RL, Q-Learning is the most classic temporal difference method that is firstly proposed by Watkins [42]. Q-Learning is one of the model-free algorithms that does not need the model information for completing learning steps and is an off-policy method that seeks to find the optimal policy in the learning process. This algorithm pursues the idea of defining Q-functions and uses the following formula for substituting the online observed data with the aim of iteratively learning Q-functions and getting the exact solution [42].

$$Q_{k+1}(s_k, a_k) = Q_k(s_k, a_k) + \alpha_k [\mathfrak{R}(s_k, a_k) + \gamma \max_{a \in \Psi} Q_{k+1}(s_{k+1}, a_k) - Q_k(s_k, a_k)] \quad (10)$$

where, k is the number of episodes and α_k is the learning rate. In this study, as all hubs are equipped with full RESs, creating a dynamic power balance between supply and demand in such an uncertain environment is a critical challenge. For this research, the multiplication of a coefficient in the difference between power supply and demand is accommodated as a reward function. Thus, the different statuses between power supply and demand are intended as the states for RL, in which imbalances between them are considered as bad conditions leading to negative rewards for the agent that guide it to adopt appropriate actions to not be in such situations again in line with maximizing its cumulative rewards. This is while adopting those actions that create a dynamic power balance opens the opportunity of receiving a positive reward for the agent if the total energy cost is reduced in comparison with the previous episode. Such interactions between the agent and environment drive the agent to learn the optimal decision-making strategy with a dynamic power balance in the presence of 100% RESs as well as the minimum energy cost for community hubs. In all studied cases for this research, all operational constraints of the integrated energy system are intended in the solving process of the optimization problem. In other words, completely satisfying all constraints is taken into account in solving the optimization problem of all cases. Thus, simulation results are obtained given the minimum value for the objective function subject to satisfying all operational constraints of the system.

3.2. Power and water network models

The optimal operation of energy hubs is conducted in the IWPG structure. To gain realistic results, considering the operational models of different power and water systems along with networks' constraints is necessary. The following equations are the operational models for various systems in the PG.

$$\sum_{h=1}^H P_{h,t}^{\{De+WUs+ELC+Ex+HtM\}} = \sum_{h=1}^H P_{h,t}^{\{BSU+FC+WS+SS+MtH\}} \quad \forall t \quad (11)$$

$$P_h^{FC, \min} \leq \left(P_{h,t}^{FC} = \phi_{h,t}^{H2,FC} \cdot \eta_{h,t}^{FC} \cdot \sigma^{Hy} \right) \leq P_h^{FC, \max} \quad \forall t, h \quad (12)$$

$$P_h^{ELC, \min} \leq \left(P_{h,t}^{ELC} = \left[\left(\phi_{h,t}^{H2,ELC} \cdot \sigma^{Hy} \right) / \eta_{h,t}^{ELC} \right] \right) \leq P_h^{ELC, \max} \quad (13)$$

$$H_{h,t}^{HSU} = H_{h,t-1}^{HSU} + \left[\left(\phi_{h,t}^{H2,ELC} - \phi_{h,t}^{H2,FC} \right) \cdot (\tau^M \cdot \epsilon^{Gas} / \lambda^{H2}) \right] \quad (14)$$

$$H_{h,t}^{HSU, \min} \leq H_{h,t}^{HSU} \leq H_{h,t}^{HSU, \max}; \quad H_{h,0}^{HSU} = H_{h,0}^{HSU} \quad (15)$$

$$P_{h,t}^{WS} = \begin{cases} 0 & 0 \leq v_t \leq v_h^{C,I}, v_h^{C,O} \leq v_t \\ P_h^{WS,Ra} \cdot \left(v_t^2 \cdot \left(\Omega_a^{WS} + \frac{\Omega_b^{WS}}{v_t} \right) + \Omega_c^{WS} \right) & v_h^{C,I} \leq v_t \leq v_h^{Ra} \\ P_h^{WS,Ra} & v_h^{Ra} \leq v_t \leq v_h^{C,O} \end{cases} \quad (16)$$

$$P_{h,t}^{SS} \leq [1 + (\kappa^{SS} - \tau^A) \cdot \delta^{SS}] \cdot [\varpi_t^{SS} \cdot \eta_h^{SS} \cdot N_h^{SS}] \quad (17)$$

$$P_h^{Ex,min} \leq P_h^{Ex} \leq P_h^{Ex,max} \quad (18)$$

$$\theta_{h,t+1}^{BSU} = \theta_{h,t}^{BSU} - \left(\eta_h^{Le,B} \cdot \theta_{h,t}^{BSU} + P_{h,t}^{BSU} + \eta_h^{CD,B} \cdot (\Pi_a^{BSU} + \Pi_b^{BSU}) \right) \cdot \Delta t \quad (19)$$

$$\partial_h^{BSU,min} \leq \left(\partial_{h,t}^{BSU} = \partial_{h,t}^{BSU} / \partial_{h,t}^{BSU,Ra} \right) \leq \partial_h^{BSU,max} \quad (20)$$

$$\Pi_a^{BSU} = -P_{h,t}^{BSU}, \Pi_b^{BSU} = P_{h,t}^{BSU}; -P_{h,t}^{BSU,Ch} \leq P_{h,t}^{BSU} \leq \bar{P}_{h,t}^{BSU,De} \quad (21)$$

$$\theta_{h,In}^{BSU} = \theta_{h,0}^{BSU}; \quad \theta_{h,End}^{BSU} \leq \theta_{h,T}^{BSU} \quad (22)$$

$$\phi_h^{BSU} = \phi_h^{BUS,IN} / (\theta_h^{BSU,Ra} \cdot N_h^{BSU,LC}) \quad (23)$$

$$\sum_{j=1}^B \text{Re}\{|V_{i,t} \cdot V_{j,t} \cdot Y_{ij}| \angle (\theta_{i,t} - \theta_{j,t} - \theta_{ij})\} = P_{i,t}^{Gen} - P_{i,t}^{Con} \quad (24)$$

$$\sum_{j=1}^B \text{Im}\{|V_{i,t} \cdot V_{j,t} \cdot Y_{ij}| \angle (\theta_{i,t} - \theta_{j,t} - \theta_{ij})\} = Q_{i,t}^{Gen} - Q_{i,t}^{Con} \quad (25)$$

$$|V_{i,t} \cdot V_{j,t} \cdot Y_{ij}| \leq S_{ij,t}^{max} \quad \forall i, j, t \quad (26)$$

$$P_i^{Gen,min} \leq P_{i,t}^{Gen} \leq P_i^{Gen,max}; \quad Q_i^{Gen,min} \leq Q_{i,t}^{Gen} \leq Q_i^{Gen,max} \quad (27)$$

$$V_i^{min} \leq V_{i,t} \leq V_i^{max}; \quad \theta_i^{min} \leq \theta_{i,t} \leq \theta_i^{max} \quad (28)$$

In the PG, balancing the power is necessary for the stability of the grid that is intended in (11). The limitation for the power consumption/generation by the EL/FC is modeled in (12)/(13). The HSU is operated subject to constraints in (14) and (15). In this respect, equation (14) models the hydrogen energy balance while equation (15) is to consider the permissible range of hydrogen energy stored in the HSU. The power outputs of WSs and SSs are respectively modeled by (16) and (17). The power exchange with the main grid is kept within the allowable range by (18). The BSU is operated subject to constraints (19) to (20) and its degradation cost is also formulated in (21) [43]. In this respect, equation (19) models the energy balance in the BSU. The standard changes of the battery's SOC are intended in equation (20). According to equation (21), the charging and discharging of the BSU can be varied in the allowable range. Equation (22) is for considering the initial and final amounts of energy stored in the BSU. The AC power flow is modeled using (24) to (28) and its linearization version is intended in this work according to the model provided in Ref. [44]. In this regard, equations (24) and (25) are for mathematically modeling the active and reactive power flow, respectively. Equation (26) keeps the amount of complex power in the standard range. The allowable variations of the active and reactive power are taken into account in equation (27). The permissible changes for the bus voltage and the related phase angle are also presented in equation (28).

The WN consists of the WD, WW, and WSU for managing water interactions in the IWPG. The operational models for the mentioned systems are as follows.

$$\sum_{h=1}^H W_{h,t}^{[De+WSch+ELC+Ex+HtM]} = \sum_{h=1}^H W_{h,t}^{[WSdis+WWdis+WD+MtH]} \quad \forall t \quad (29)$$

$$P_{h,t}^{WUs} = P_{h,t}^{WSU,Ch} + P_{h,t}^{WD} + P_{h,t}^{WW,Dis} \quad \forall h, t \quad (30)$$

$$W_h^{Ex,min} \leq W_{h,t}^{Ex} \leq W_h^{Ex,max} \quad \forall h, t \quad (31)$$

$$P_{h,t}^{WSU,Ch} = \left[(\Upsilon_{h,t}^{WS} + \Upsilon_{h,t-1}^{WS} + \Upsilon_h^{A,WS}) \cdot \left(\frac{W_{h,t}^{WSch} \cdot \psi^{Wa} \cdot \gamma^{Wa}}{\eta_h^{WP} \cdot (7.2 \times 10^{+6})} \right) \right] \quad (32)$$

$$0 \leq (\Upsilon_{h,t}^{WS} - \Upsilon_{h,t-1}^{WS} - \left[(W_{h,t}^{WSdis} - W_{h,t}^{WSch}) / \mu^{WS} \right]) \leq \Upsilon_h^{WS,max} \quad (33)$$

$$W_{h,t}^{WSdis} \leq \chi_{h,t}^{WSdis} \cdot W_h^{WSdis,max}; \quad W_{h,t}^{WSch} \leq \chi_{h,t}^{WSch} \cdot W_h^{WSch,max} \quad (34)$$

$$\chi_{h,t}^{WSdis} + \chi_{h,t}^{WSch} \leq 1 \quad \forall h, t \quad (35)$$

$$0 \leq (P_{h,t}^{WD} = \eta_h^{WD} \cdot W_{h,t}^{WD}) \leq P_h^{WD,max} \quad \forall h, t \quad (36)$$

$$P_{h,t}^{WW,Dis} = \left[(W_{h,t}^{WWdis} \cdot \Upsilon_h^{WW}) \cdot \left(\frac{\psi^{Wa} \cdot \gamma^{Wa}}{\eta_h^{WWp} \cdot (3.6 \times 10^{+6})} \right) \right] \quad \forall h, t \quad (37)$$

The water balance of the WN is established using (29). The total consumed power in the WN is stated by (30). The water exchange constraint is intended in (31). The consumed power by the WSU in the charging state is modeled by (32). Constraints (33) to (35) formulate the operational limitations of the WSU. In this respect, the balance constraint for the stored water in the WSU is stated in equation (33). The limitations of water charging and discharging in the WSU are presented in equation (34). As the WSU cannot be simultaneously operated in both charging and discharging modes, constraint (35) mathematically models this point by defining two binary variables for the charging and discharging states. The consumed power by the WW and WD are respectively computed by (36) and (37).

3.3. Twofold ancillary service (TAS) framework

This work studies the operation of community energy hubs that are planned to utilize 100% RESs for completely producing cost-effective and pollutant-zero energy. Such an uncertain environment requires strong flexibility schemes and ancillary services to make the system confident from reliably and continuously serving energy. To overcome this challenge, this work develops an innovative TAS framework that benefits both the transactive energy technology and demand response programs to procure the required flexibility for renewable-dominant hubs. In this respect, transactive energy is the first leverage of the proposed framework that establishes free power and water sharing among hubs in the PWLEM. All hubs can benefit from transactive energy trading to manage their energy interactions to improve their energy-serving conditions and maximizing economic achievements. The power and water sharing among hubs can be carried out in the PWLEM based on the following transactive energy constraints.

$$\chi W_{h,t}^{HtM} + \chi W_{h,t}^{MtH} \leq 1; \quad \chi E_{h,t}^{HtM} + \chi E_{h,t}^{MtH} \leq 1 \quad \forall h, t \quad (38)$$

$$W_{h,t}^{MtH} \leq \phi^{Wa} \cdot \chi W_{h,t}^{MtH}; \quad P_{h,t}^{MtH} \leq \phi^{EL} \cdot \chi E_{h,t}^{MtH} \quad \forall h, t \quad (39)$$

$$W_{h,t}^{HtM} \leq \phi^{Wa} \cdot \chi W_{h,t}^{HtM}; \quad P_{h,t}^{HtM} \leq \phi^{EL} \cdot \chi E_{h,t}^{HtM} \quad \forall h, t \quad (40)$$

$$\sum_h P_{h,t}^{HtM} = \sum_h P_{h,t}^{MtH}; \quad \sum_h W_{h,t}^{HtM} = \sum_h W_{h,t}^{MtH} \quad \forall t \quad (41)$$

$$\sum_t (\rho_t^{W,Ex} \cdot W_{h,t}^{HtM}) = \sum_t (\rho_t^{W,Ex} \cdot W_{h,t}^{MtH}) \quad \forall h \quad (42)$$

$$\sum_t (\rho_t^{E,Ex} \cdot P_{h,t}^{HtM}) = \sum_t (\rho_t^{E,Ex} \cdot P_{h,t}^{MtH}) \quad \forall h \quad (43)$$

In (38), the state of the water and power exchange in the PWLEM is modeled. The water and power sharing limitations in the transactive energy market are considered in (39) and (40). The energy balance of the market is created by (41). The economic balance for hubs is also intended by (42) and (43).

The second leverage of the TAS framework is to adopt demand response programs for flexibility by intending the elasticity of loads in shifting and even curtailing. Given the mentioned attributes of the

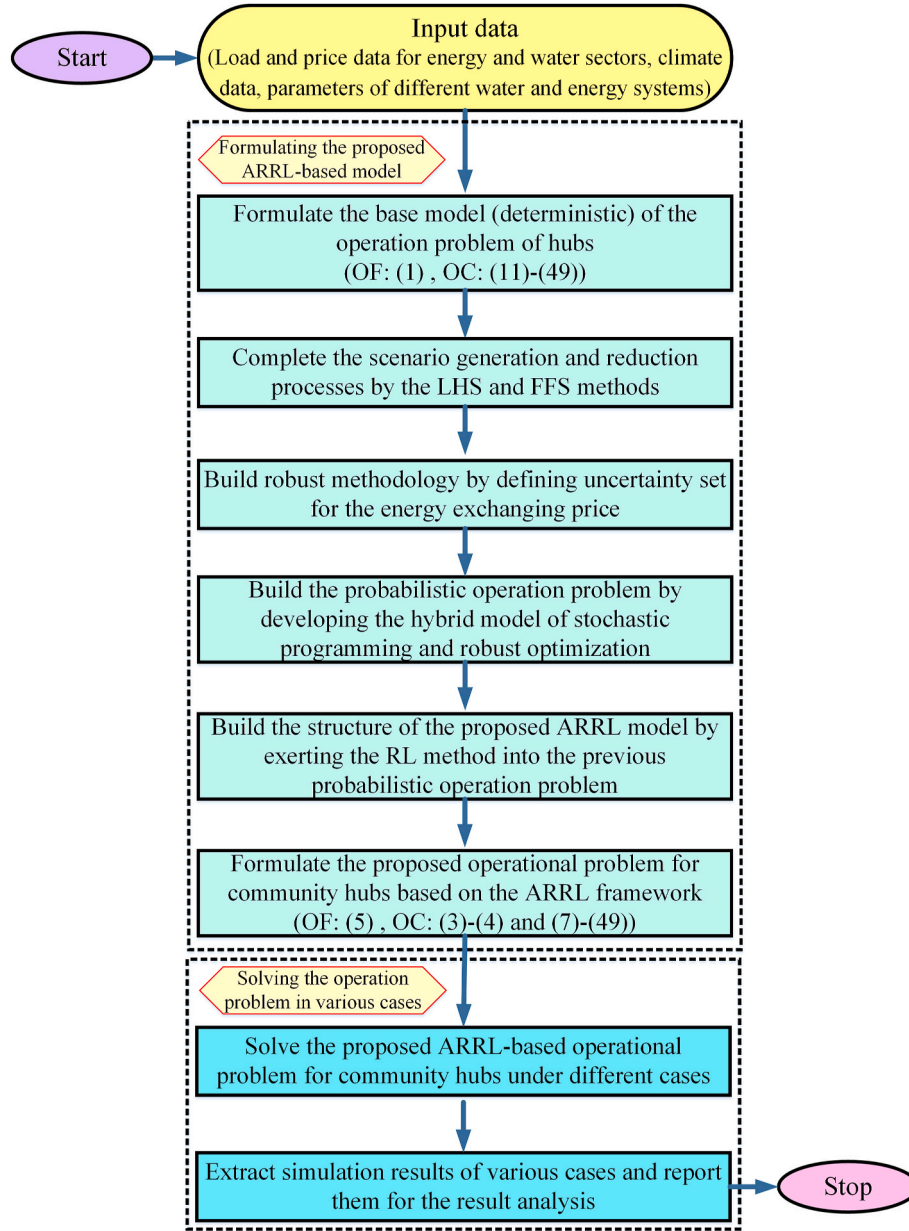


Fig. 2. The methodology flowchart for the operation of community hubs.

dispensable loads and for enhancing the flexibility of hubs [45], the power and water management of the demand side is accomplished by adopting load shifting and curtailing programs as follows.

$$\text{Cos } t_t^{EM} = \underbrace{\left(\theta_h^{ILP1} \cdot \left(\mathbb{R}_{h,t}^{ILP} \right)^2 + \theta_h^{ILP2} \cdot \mathbb{R}_{h,t}^{ILP} \right)}_{C_t^{ILP}} + \underbrace{\left(\rho_t^{SHP} \cdot \mathbb{R}_{h,t}^{SHP+} / 2 \right)}_{C_t^{SHP}} \quad (44)$$

$$\mathbb{R}_{h,t}^{De} = \mathbb{R}_{h,t}^{De,Fo} + \mathbb{R}_{h,t}^{SHP} - \mathbb{R}_{h,t}^{ILP} \quad \mathbb{R} \in \{P, W\} \quad (45)$$

$$\mathbb{R}_{h,t}^{ILP,min} \leq \mathbb{R}_{h,t}^{ILP} \leq \mathbb{R}_{h,t}^{ILP,max} \quad \forall h, t \quad (46)$$

$$\mathbb{R}_{h,t}^{De,Do} \leq \mathbb{R}_{h,t}^{De} - \mathbb{R}_{h,t}^{ILP} \leq \mathbb{R}_{h,t}^{De,Up} \quad \forall h, t \quad (47)$$

$$\mathbb{R}_{h,t}^{SHP,min} \leq \mathbb{R}_{h,t}^{SHP} \leq \mathbb{R}_{h,t}^{SHP,max} \quad \forall h, t \quad (48)$$

$$\sum_{t=1}^T \mathbb{R}_{h,t}^{SHP} = 0 \quad \forall h \quad (49)$$

The first and second terms of (44) denote the cost of curtailing and shifting programs. The limitations for the load curtailing are accommodated by (45) to (47). The constraints for the shifting program are also intended by (48) and (49). To clearly state the problem formulation and solving processes of this work, the methodology flowchart of this research is illustrated in Fig. 2.

As stated in Fig. 2, the problem analysis is started by inputting the

Table 2

Case studies in the operation of energy hubs.

Case studies	Model characteristics				Objective function	Operational Constraints
	TAS	HRS	RL	ARRL		
Case I	✓	×	×	×	(1)	(11) To (49)
Case II	✓	✓	×	×	(5)	(3) To (4), (11) to (49)
Case III	✓	×	✓	×	(1)	(7) To (49)
Case IV	✓	✓	✓	✓	(5)	(3) To (4), (7) to (49)

required data for the system model which includes the load and price data for the energy and water sectors, data for modeling the energy grid, wind speed and solar irradiation as climate data, and other required parameters for modeling each of the water and energy systems. In the next step, the basic or deterministic version of the optimization problem is shaped that considers the system model without the uncertainty quantification. For the probabilistic analysis of the system, first, the stochastic process is carried out by applying the LHS and FFS approaches. Second, robust optimization is applied for the problem formulation to make the building of a probabilistic operation problem using the HRS method possible in the next step. In the two last parts of the problem formulation, the proposed ARRL model is manipulated by the RL approach and then the final structure of the optimization problem is constructed by applying this model to the previous probabilistic operation problem. Afterward, the ARRL-based problem is solved by considering various cases and the results are reported for scrutinizing the different achievements of this research.

4. Results and discussion

One of the main goals of this research is to unlock the challenge of finding the optimal decision-making strategy for the optimal operation of energy hubs by developing the machine learning-based ARRL model. To examine the effectiveness of the proposed model, four case studies are considered with different properties. Table 2 clearly indicates various characteristics of the studied cases.

According to Table 2, Case I intends the primary state of the operation model without considering the RL and robust algorithms, so it is recognized as the basic model of the study. Case II analyzes the effects of applying robust stochastic optimization to the basic model to model intermittences in an intense deregulated environment. However, this case does not include the RL for learning the optimal strategy. This is while Case III evaluates the application of the RL by exerting the Q-learning to the basic model without considering the robustness of the system by not using robust stochastic optimization. However, to simultaneously benefit from the advantages of both RL and HRS, Case IV uses the proposed ARRL model to explore its effectiveness in comparison with the previous cases. Indeed, this case assesses the capability of the developed ARRL model in detecting the optimal decision-making strategy for the exploitation of energy hubs with 100% RESs.

Detecting the optimal decision-making strategy is one of the prominent goals that is pursued in this work by proposing a novel ARRL model. Another key goal is to produce 100% clean power by deploying RESs and other supportive devices. To this end, WSs and SSs are employed along with the BSU to offer a confident process for supplying clean power [45]. This confidence is further improved by adopting an energy conversion process that creates energy interactions in the IWPG

and consists of the EL for hydrogen generation, the HSU for hydrogen storage, and the FC for power reproduction [46]. The WSU, WW, and WD are water systems that enable the WN for stable operation [47]. To scrutinize power and water interactions in the IWPG, the modified IEEE 33-bus test system is taken into account according to the schematic shown in Fig. 3.

The optimization problem is solved for the optimal operation of energy hubs in the four described case studies. For these cases, the total operating costs are respectively gained \$57304.03, \$61501.69, \$43286.57, and \$47545.78. Given the costs for different cases, Case II imposes more cost for hubs in comparison with Case I due to procuring robustness against the fluctuations of RESs in the system. Also, Case IV has more operating cost than Case III due to considering uncertainties for realistic studying the energy structure. Indeed, when uncertain behaviors of different uncertain sources are modeled in the system to give a most realistic state of modeling for the studied system, Cases II and IV are exposed to more operation costs to obtain sufficient robustness against intermittences of uncertainties under the probabilistic assessment of the system. On the other hand, using the RL-based method for learning the optimal decision-making strategy in Cases III and IV has resulted in cost reduction in comparison with Cases I and II without RL methodology. On the other hand, comparing the results of Case IV with the ARRL model and Case II without applying the mentioned model indicates that benefiting the capabilities of the ARRL model has achieved 22.7% cost savings for Case IV than Case II. In other words, the usage of RL in detecting optimal set points of different units has empowered the proposed strategy to use the purposeful optimal policy detection process in RL to find the optimal policy for the optimal operation of community energy hubs. Indeed, RL provides the advantages of deploying an agent to discover optimal passes based on the defined rewards in line with the objective function and constraints. According to the conducted comparison for the results of Cases II and IV, such advantages of RL not only have brought remarkable cost savings for Case IV in comparison with Case II but also have procured optimal policy for the optimal exploitation of power and water systems in community hubs. In this respect, all hubs use RESs and the BSU to realize full power production, which their variations are depicted in Fig. 4.

Fig. 4 indicates the satisfactory level of WSs output in the early morning when the system relies on wind power as the only RESs. However, this status no longer remained by gradually increasing the intensity of solar radiation and substantially raising the output of SSs. Such a trend has led to the maximum clean power generation in peak times. However, although the evening time period has suitable wind power production, the last 3 h of the night have the minimum carbon-free power production during the day. Due to this, the BSU experiences a high level of discharging in the mentioned hours to support hubs for dynamically balancing power and maintaining the stability of the system. Indeed, WSs and SSs can deliver clean power at most times of the day, especially in peak times when the outputs of SSs are the maximum compared to other hours. However, their intermittent power production has led to forcing the system to not be operated without the BSU to keep the continuous power supply for consumers. On the other hand, all hubs can benefit from the power-sharing possibility to flexibly manage power interactions in different time periods. In this respect, energy hubs can freely trade power with each other in the PWLEM based on the transactive energy technology and they also can have limited power exchange with the main grid. Fig. 5 illustrates the optimal power sharing for community hubs.

In Fig. 5, energy hubs have bought more power from the power grid in the time period 1–8 a.m. to effectively utilize the opportunity of buying cheaper power. However, the noon hours with the maximum RESs output and a higher power price have provided an appropriate condition for hubs to sell their extra power to the upstream grid to maximize their profit. This is while from 8 to 12 p.m., energy hubs have received electrical energy from the main grid to cover a portion of the power shortage in the system. Moreover, all hubs have availed the

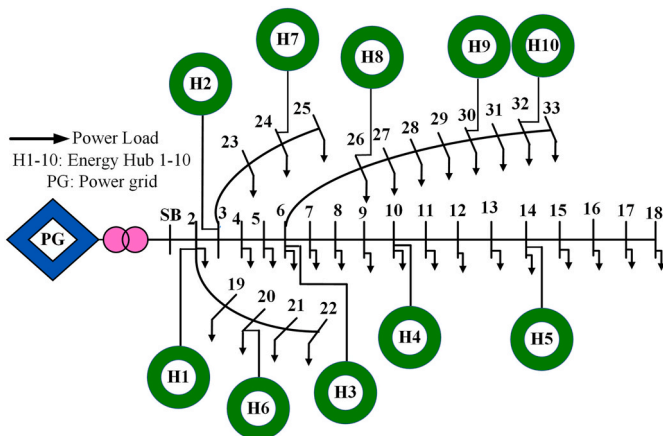


Fig. 3. Community hubs in the modified IEEE 33-bus test system.

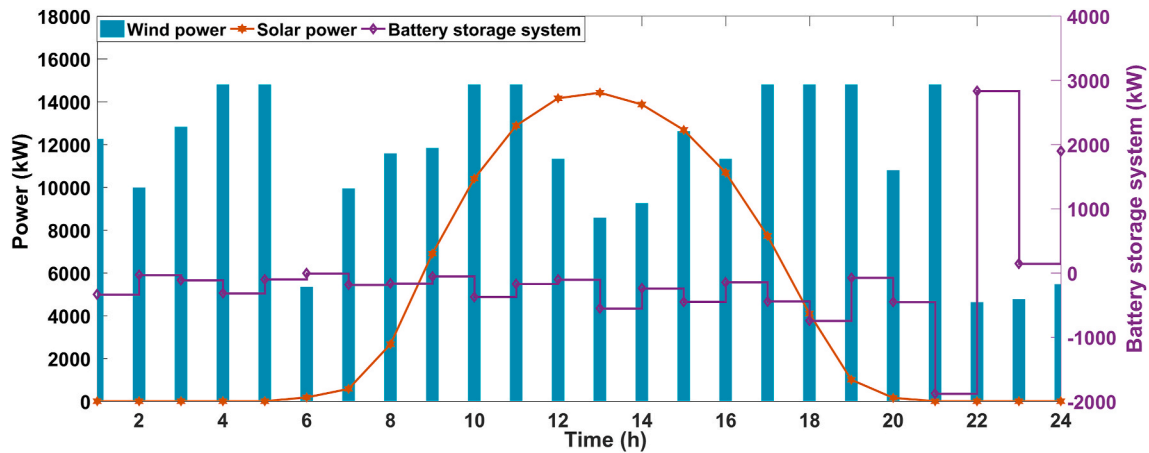


Fig. 4. Produced power by RESs along with power changes in the BUS.

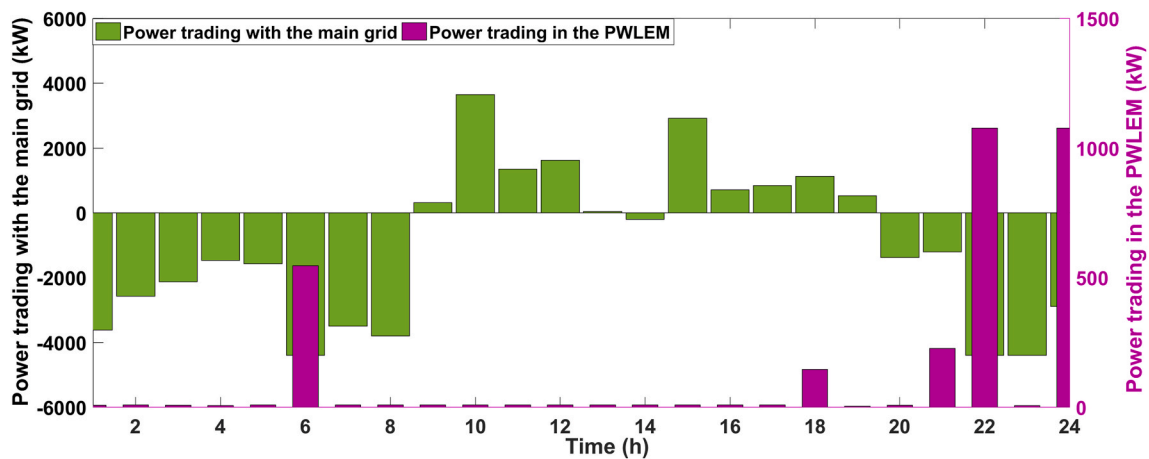


Fig. 5. The optimal power sharing for community hubs.

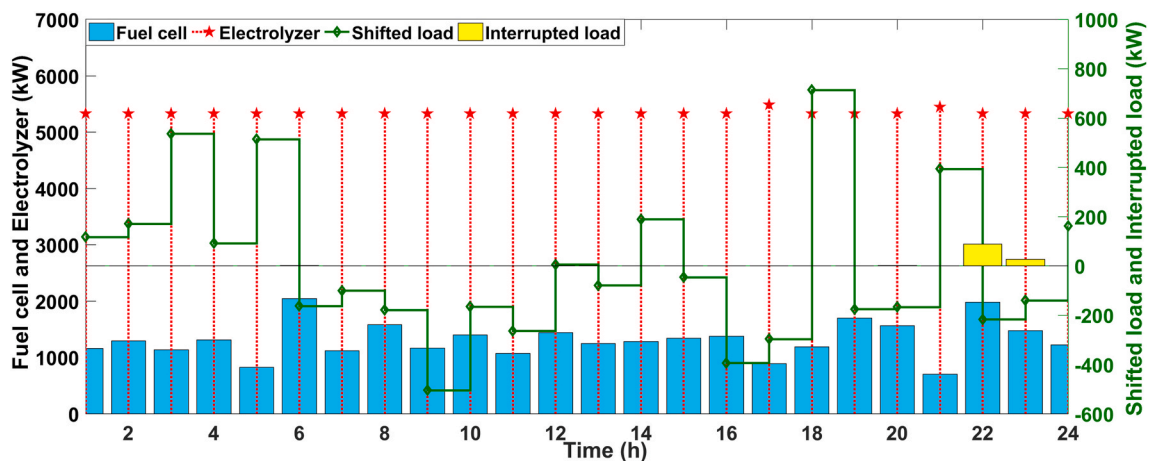


Fig. 6. Outputs of the FC and EL along with the demand response programs.

participation opportunity in the transactive energy market to freely trade power to gain more profit while keeping the power balance. According to Fig. 5, transactive energy trading is used in critical times of the day when energy hubs suffer from lower power generation and need other reliable ways for balancing power. Indeed, Fig. 5 identifies that energy trading is one of the necessary ways to establish adequate flexibility for the renewable-based hubs. Herein, both the potential of the

grid and community hubs are available for developing an energy trading strategy aiming at flexibility improvement. Energy sharing is done among hubs in the PWLEM and the local level when the generation units of energy hubs are in the suitable status for power production and in hours with a lower output for renewables, the potential of the grid is intended for power transition between energy hubs and the upstream network. In this regard, community hubs are also benefited from energy

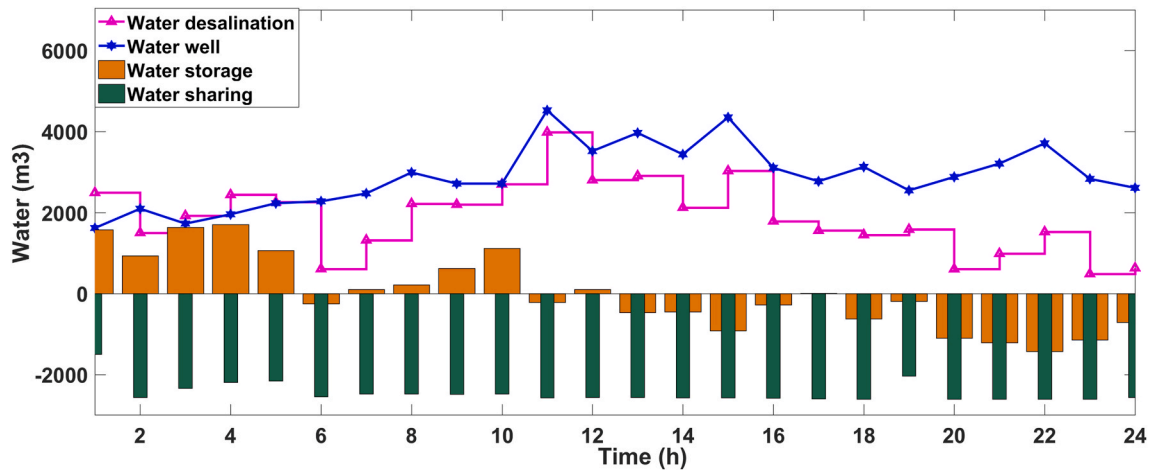


Fig. 7. Optimal set points of the WN devices.

conversion facilities along with the demand response programs as other ways to improve their flexibility in the presence of 100% RESs. The outputs of the FC and EL along with the demand response programs are portrayed in Fig. 6.

As obvious from Fig. 6, the FC has mostly played a supportive role in the IWPG to keep its power and water balance. Indeed, this unit has generated more power at 6 and 8 a.m. along with the last hours of the day when multi-energy hubs face the limitation of adequate power production in the system. This is made possible for energy hubs by receiving the required hydrogen molar generated by the EL system. On the other hand, the dispensable loads allow the system to use their elasticity properties for shifting the power demand from time periods with lower available power to power-rich hours under the demand response program. On another side of managing the demand side, the system is allowed to utilize the interrupted load program for curtailing a limited range of the demand only at 22 and 23 p.m. as emergency times. The information in Fig. 6 also specifies that when the energy structure is equipped with 100% renewables, energy conversion can be taken into account as another effective way of uninterrupted energy delivery. This figure effectively indicates how the coordinated operation of different energy conversion systems can support hubs to confidently deploy the high potential of renewables while satisfying power demand. In addition to the operation of the PG, the optimal operation of the IWPG is accompanied by determining the optimal set points of the WN devices that are demonstrated in Fig. 7.

Fig. 7 denotes that the WSU is in the charging state in the early morning hours to benefit from the lower water price for water purchasing and storing. This is while by growing the water consumption from 10 a.m., the WSU has supported the WN in water serving by switching to the discharging state. The WW and WD are two typical

water production systems that have lower water output in the early morning with lower water consumption. However, their water generation range is increased from 10 a.m. when the water demand is raised in peak hours. The water sharing is also intended as another flexibility option for the WN that is availed during the day for establishing a water balance between the water supply and demand. Indeed, similar to the power sector, Fig. 7 also clearly illustrates the undeniable role of water trading in maintaining the sustainability of energy hubs in water supply. In this respect, the coordinated operation of water systems not only enables hubs to meet water demand but also allows for effectively managing power and water interactions in the interdependent structure of the system. For the optimal operation of the IWPG, this work proposes an ARRL model for community hubs with 100% RESs to procure a sufficient degree of robustness against the stochastic volatilities of RESs. The operation costs of energy hubs in different robustness conditions are illustrated in Fig. 8.

Fig. 8 clearly shows that by increasing the budget of uncertainty, the operation cost is subsequently increased due to raising the degree of robustness that comes from intending the worst possible changes of the uncertain parameter. On the other hand, the energy cost is also increased by upsurging the deviation from the forecasted price. This is because the system can be more robust in the larger deviation period of the uncertain parameter that imposes higher costs. In this respect, this is up to the decision-maker to select the satisfactory trade-off between the cost and robustness given the type of the problem. In this respect, Fig. 8 clarifies that uncertainty quantification is prominent for assessing the economic viability of the operation model. It also states that obtaining realistic results can be achieved when the realistic conditions of the system are taken into account by effectively modeling the stochastic changes of uncertain sources by the capable uncertain modeling

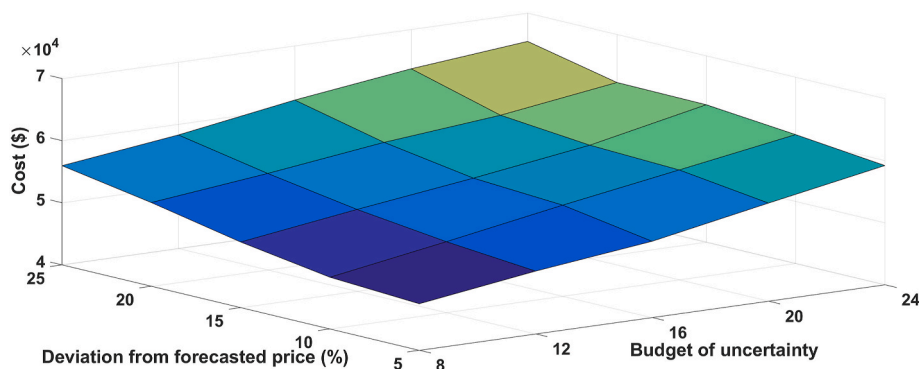


Fig. 8. Operation costs of energy hubs in different robustness conditions.

approach.

5. Conclusion

In this article, an innovative ARRL model was proposed to detect the optimal decision-making strategy for the optimal operation of renewable-dominant energy hubs in the IWPG. By equipping to 100% RESs, energy hubs were targeted to use the advantages of fully carbon-free power generation in line with zero-carbon emissions plans. However, as reaching this goal brings challenges in reliable power supply due to the stochastic intermittences of uncertain parameters, the HRS technique was developed to adaptively model uncertainties by intending the diverse behaviors in uncertain changes. The TAS framework was proposed to allow community hubs to flexibly interact in the intense deregulated environment by simultaneously benefiting from the trans-active energy trading in the PWLEM as well as demand response programs by using the elasticity attributes of dispensable loads. The effectiveness of the proposed RL-based model was examined by analyzing four case studies considering the modified IEEE-33 bus test system. The proposed model in Case IV could render cost-effective conditions by procuring a 17.03% cost saving in comparison with the basic model in Case I. Furthermore, analyzing numerical results proved that the system imposed 8.96% more cost when energy hubs were operated under uncertainty modeling in comparison with the basic model of the operation without the consideration of uncertainties. Indeed, the proposed ARRL model can not only learn the optimal decision-making strategy for the optimal exploitation of energy hubs with 100% RESs but also can properly model uncertainties by exerting the developed HRS method leading to the realistic modeling of hubs as well as facilitating the achievement of techno-economic-environmental goals.

This work scrutinizes the optimal exploitation of community energy hubs by developing a new decision-making strategy in the presence of 100% RESs to produce full clean energy. The simulation results indicated multi-dimensional achievements of the proposed model particularly in terms of economic, environmental, and technical. However, the lack of a real-world case with hybrid power and water systems, 100% RESs, and other similar conditions limits the complete assessment of the model's capability in practice. As the main focus of this work is to model the coordinated operation of water and power systems to assess the role of water units in improving the flexibility of the power grid, the operation of different water systems is considered in energy interactions without considering the water network model. For a detailed analysis of the effects of the water sector on the electric power system dispatch, the water network model will also be intended in the related structure and assessment in future work.

CRedit authorship contribution statement

Mohammadreza Daneshvar: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Ahmadian:** Writing – review & editing, Investigation, Conceptualization. **Ali Diabat:** Writing – review & editing, Investigation, Conceptualization. **Ali Elkamel:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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